

Rooftop solar for poverty alleviation

A detailed white paper prepared by The Infravision Foundation proposes a Central government-sponsored scheme that fundamentally looks at rooftop solar photovoltaics (RTPV) as the next trailblazing income-enhancing opportunity, targeted largely at those at the bottom of the pyramid, and largely in rural areas. It has the potential to be a hugely impactful intervention to aid poverty alleviation by converting free sunlight falling on rooftops and courtyards into income for households. For instance, in China, RTPV is one of the identified 10 initiatives rolled out by the government to lift rural households out of poverty. Last year, 100 per cent solarisation in the village of Modhera in Mehsana district, Gujarat, has demonstrated the transformative potential of rooftop solar in an Indian village.

For a variety of reasons, rooftop solar in India has not yet been the success it was planned to be. That is a pity. A scheme, of the type suggested, seeks to breathe fresh life into this area, targeting not just "green" but, even more importantly, is an innovative income-generating scheme. It could well be called *Sooraj Se Rozgaari* (or, earnings from sunlight).

The proposed scheme assumes a customer mix (see table).

According to this customer configuration, the Central government gets involved through the Indian Renewable Energy Development Agency (IREDA). With Central-government financial assistance, the IREDA procures and deploys rooftop modules and assisted infrastructure via its extended arm of State Renewable Energy Development Agencies (SRDAs). Bulk procurement by the IREDA is expected to reduce capital costs by providing high-volume orders to module manufacturers. SRDAs secure state regulatory approvals on benchmark costs of RTPV, including state-specific grid-installation charges and a fair developer margin. Benchmark costs reflect locational and services markup on top of the capital costs discovered by the IREDA.

A new generation of local "developers" has to market the scheme and source consumer interest

and, thereafter, approach SRDAs with the request for necessary modules and infrastructure. They then do the installation at consumers' premises and are responsible for maintenance for at least 15 years. They are paid a fixed installation fee and yearly maintenance fees from SRDAs for their services.

Consumers may have to share part of the benchmark costs, depending upon which category they fit in — from low-income households to institutional or small businesses. They will have to agree upon their share of self-use of the electricity generated from RTPV and the balance that will be considered as sold to SRDAs (unless bought by the discom), for which they will be suitably compensated.

The SRDAs aggregate the balance electricity from all participating consumers, and are encouraged to maximise revenue by sale of this power to other discoms, large consumers, and power exchanges. Price-arbitrage opportunities, that is the difference between the compensation paid to participating consumers and the realised sale price for electricity sold to other parties, will ultimately determine the net financial assistance support by the Central government.

Currently, the rooftop-subsidy programmes run by the Ministry of New and Renewable Energy require consumers to bear about 60 per cent of the costs. For the suggested programme, it is recommended that consumers be grouped into three distinct categories: (i) low-income households; (ii) social, institutional, and small businesses such as municipal schools and district hospitals, non-government organisations, farmer producer organisations, small and medium enterprises (SMEs) or micro-SME businesses, and (iii) regular-income households. It is proposed that low-income households get full capital grants, i.e they do not have to pay for any share of the costs for RTPV installation and main-

CUSTOMER MIX UNDER THE SCHEME

	Units	Low-income households	Regular income households	Institutional/ MSME	Total
No. of customers	Lakhs	100	20	8	128
System capacity per customer	kW	1	3	5	
Total installed capacity	GW	10	6	4	20
Proposed subsidy	%	100	40	20	

tenance. Social/institutional/small businesses bear 80 per cent of the costs, while the households with regular incomes bear 60 per cent, which is in accordance with current practice.

Developers have the responsibility to market the programme benefits to consumers and agglomerate the demand. Once the consumer puts in his/her contribution (in accordance with set guidelines), the developer installs the requisite RTPV on consumers' premises. The developer gets paid a fixed installation fee and yearly maintenance fees from SRDAs for its services. This naturally creates a new set of local energy-sector entrepreneurs, in line with the government's vision of creating such opportunities at district level.

The proposed centralised scheme is expected to add 20 Gw of residential RTPV capacity over a five-year period. The "NET" Central sector budgetary support required will be approximately ₹19,950 crore over a 15-year period assumed for the scheme.

Above all, a low-income household with limited roof space stands to benefit (based on the final interplay of assumptions) anything from ₹3,500 to ₹6,000 per annum from the "free" sunlight on its abode. Combined with the "green effect", such a poverty-alleviating opportunity can surely not be left unexploited.

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